

MID TERM EXAMINATION

Paper: Mid Term Examination - Set B | Duration: 3 Hours | Max Marks: 20 | Date: 13-03-2026

Total Questions: 10 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQS

1. What are the three fundamental uses of architecture documentation as outlined in the text? [2 Marks]
 - a) Education, communication among stakeholders, and basis for system analysis and construction.
 - b) Project planning, risk management, and client presentations.
 - c) Performance monitoring, security audits, and budget allocation.
 - d) Marketing, legal compliance, and training new employees.
2. Which of the following is explicitly listed as an architectural quality attribute in the text? [2 Marks]
 - a) Time to Market
 - b) Reliability
 - c) Conceptual Integrity
 - d) Marketability
3. According to the definition provided, what does the software architecture of a program or computing system comprise? [2 Marks]
 - a) Only software elements and their provided services.
 - b) Software elements, their externally visible properties, and the relationships among them.
 - c) The source code files, database schemas, and user interface designs.
 - d) Business requirements, technical specifications, and development timelines.
4. The text states that an architecture is influenced by several factors. Which of the following best describes the influence of "The Developing Organization"? [2 Marks]
 - a) The specific preferences and coding styles of individual developers.
 - b) Immediate business investment, long-term business investment, and organizational structure.
 - c) The availability of open-source libraries and frameworks.
 - d) External market trends and competitor analysis.
5. What is the primary distinction between a "Reference Model" and a "Reference Architecture" as defined in the source text? [2 Marks]
 - a) A Reference Model divides functionality, while a Reference Architecture maps that functionality onto software elements.
 - b) A Reference Model describes element types, while a Reference Architecture describes quality attributes.
 - c) A Reference Model focuses on runtime components, while a Reference Architecture focuses on compile-time modules.
 - d) A Reference Model is a formal specification, while a Reference Architecture is an informal guideline.
6. When discussing Component-and-Connector (C&C) Views, what are "ports" described as? [2 Marks]
 - a) The internal data structures used by a component.

- b) Points of potential interaction of a component with its environment, usually having an explicit type.
- c) The communication protocols linking components together.
- d) The user interfaces provided by the system's components.

7. According to the text, what is a key reason why architecture reconstruction is described as an "interpretive, interactive, and iterative process" and not automatic?

[2 Marks]

- a) Because architecture documentation is always complete and up-to-date.
- b) Architectural constructs are not explicitly represented in the source code, requiring inverse mappings and architectural insight.
- c) The process only involves technical tools like grep and sed, which are not automated.
- d) It primarily focuses on identifying elements for reuse, which is a manual task.

8. Which of the following is NOT explicitly listed as an advantage of Object-Oriented Architecture in the provided text?

[2 Marks]

- a) Improves testability through encapsulation.
- b) Reduces development time and cost.
- c) Guarantees perfect security and fault tolerance.
- d) Provides reusability through polymorphism and abstraction.

9. In the context of the Attribute-Driven Design (ADD) steps, what is the purpose of "Verifying and refining use cases and quality scenarios and making them constraints for the child modules" in Step 2.e?

[2 Marks]

- a) To immediately allocate all remaining functionality to the lowest-level modules.
- b) To ensure that nothing important was overlooked and to prepare the child modules for further decomposition or implementation.
- c) To finalize the system's external interfaces without any possibility of further change.
- d) To replace the initial architectural drivers with new ones for the child modules.

10. The "Software Architecture Business Cycle" section explains how architectures are influenced by various factors. Considering the statement "Requirements make explicit some?but only some?of the desired properties" and the subsequent points about stakeholders, which of the following best captures the architects' crucial role in relation to stakeholders and requirements?

[2 Marks]

- a) Architects should only implement properties explicitly stated in the requirements document.
- b) Architects must identify and actively engage stakeholders early to understand real constraints, manage expectations, negotiate, and resolve conflicts.
- c) Architects should unilaterally decide on system priorities, as stakeholders' concerns may be contradictory.
- d) Architects primarily need technical skills to translate all stakeholder demands into direct architectural decisions.

--- End of Question Paper ---